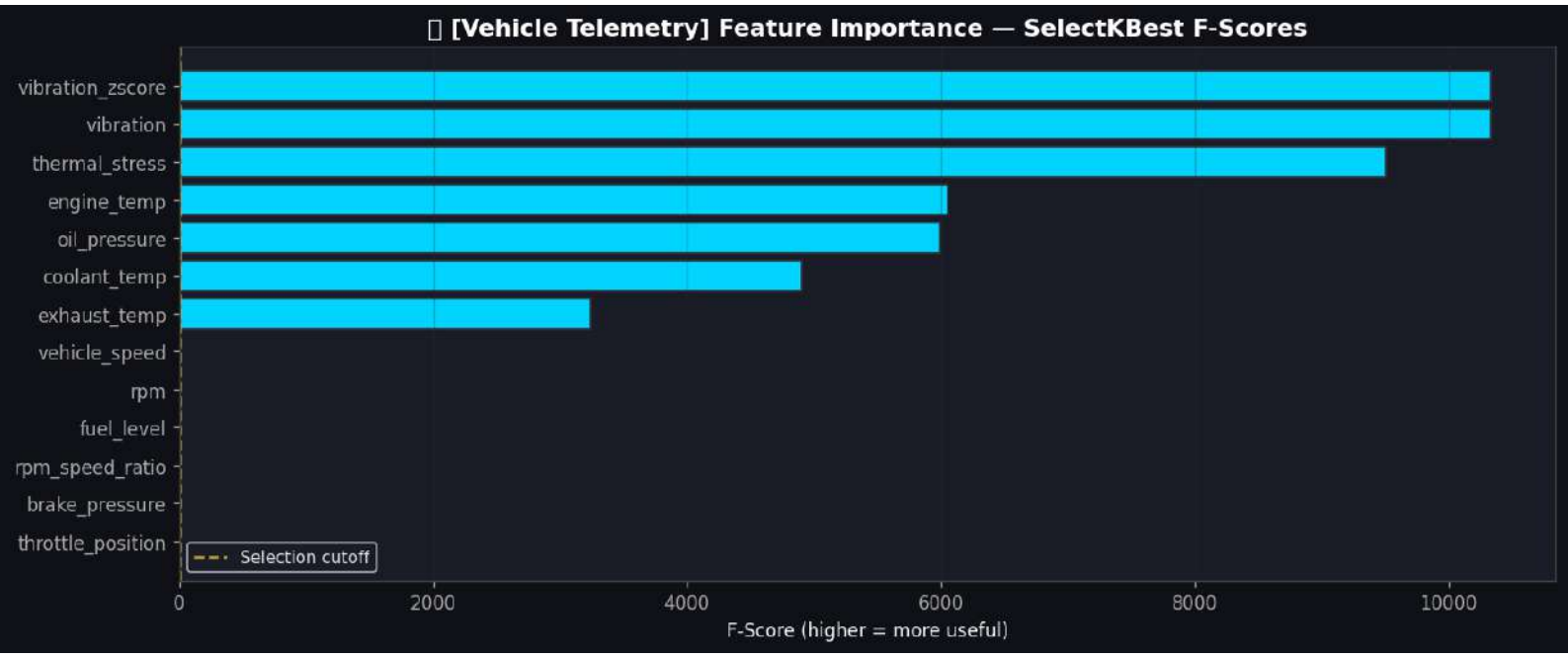
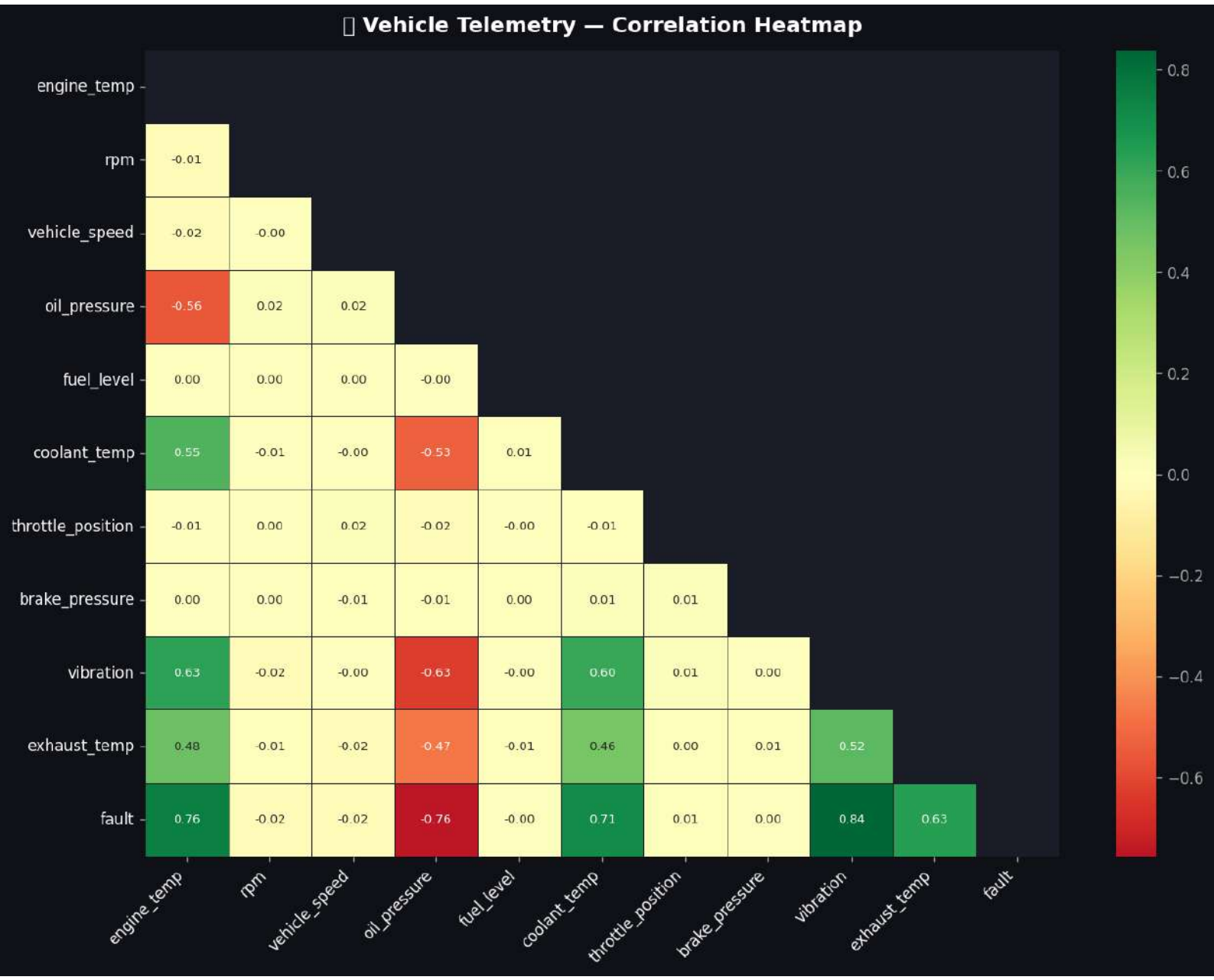
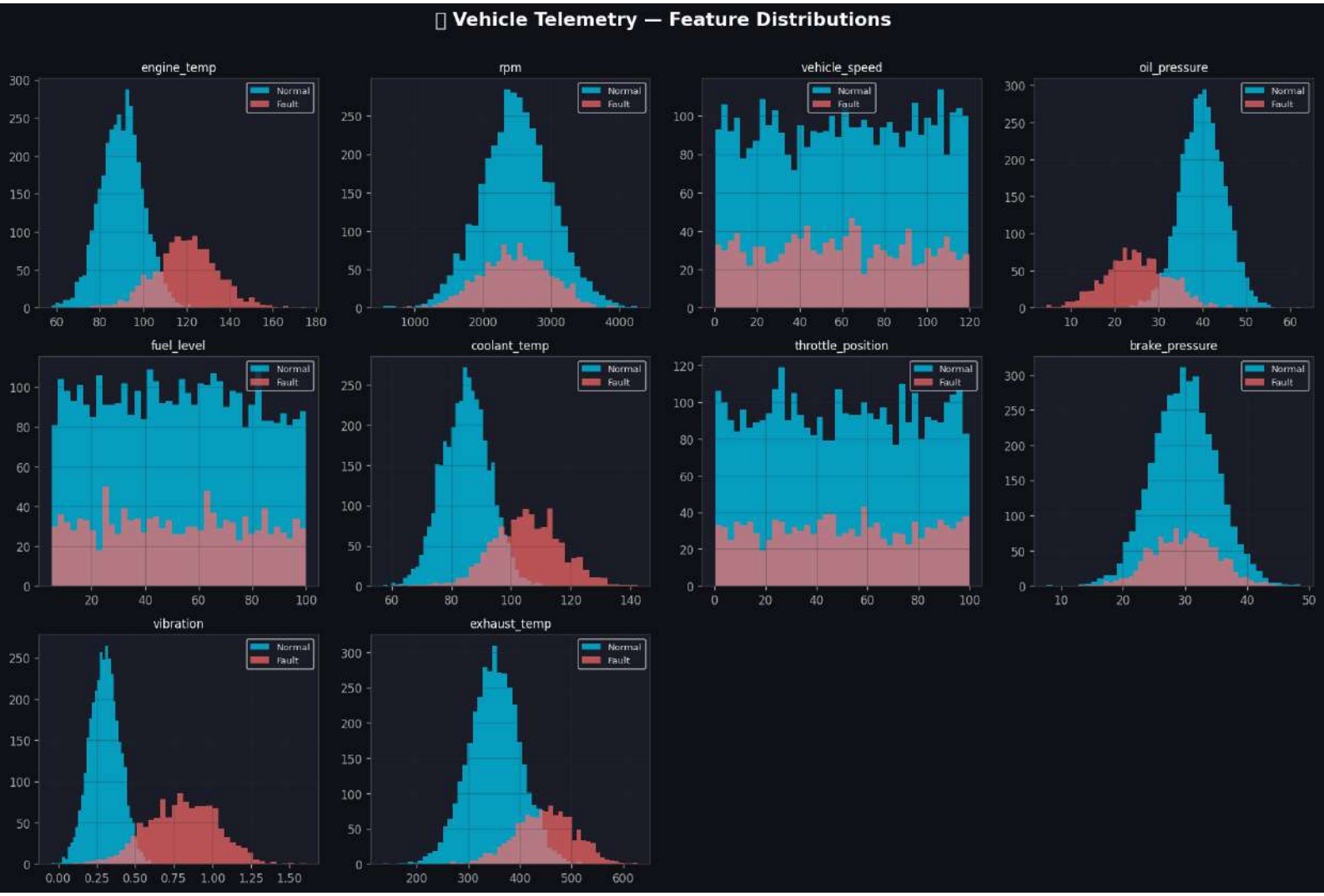
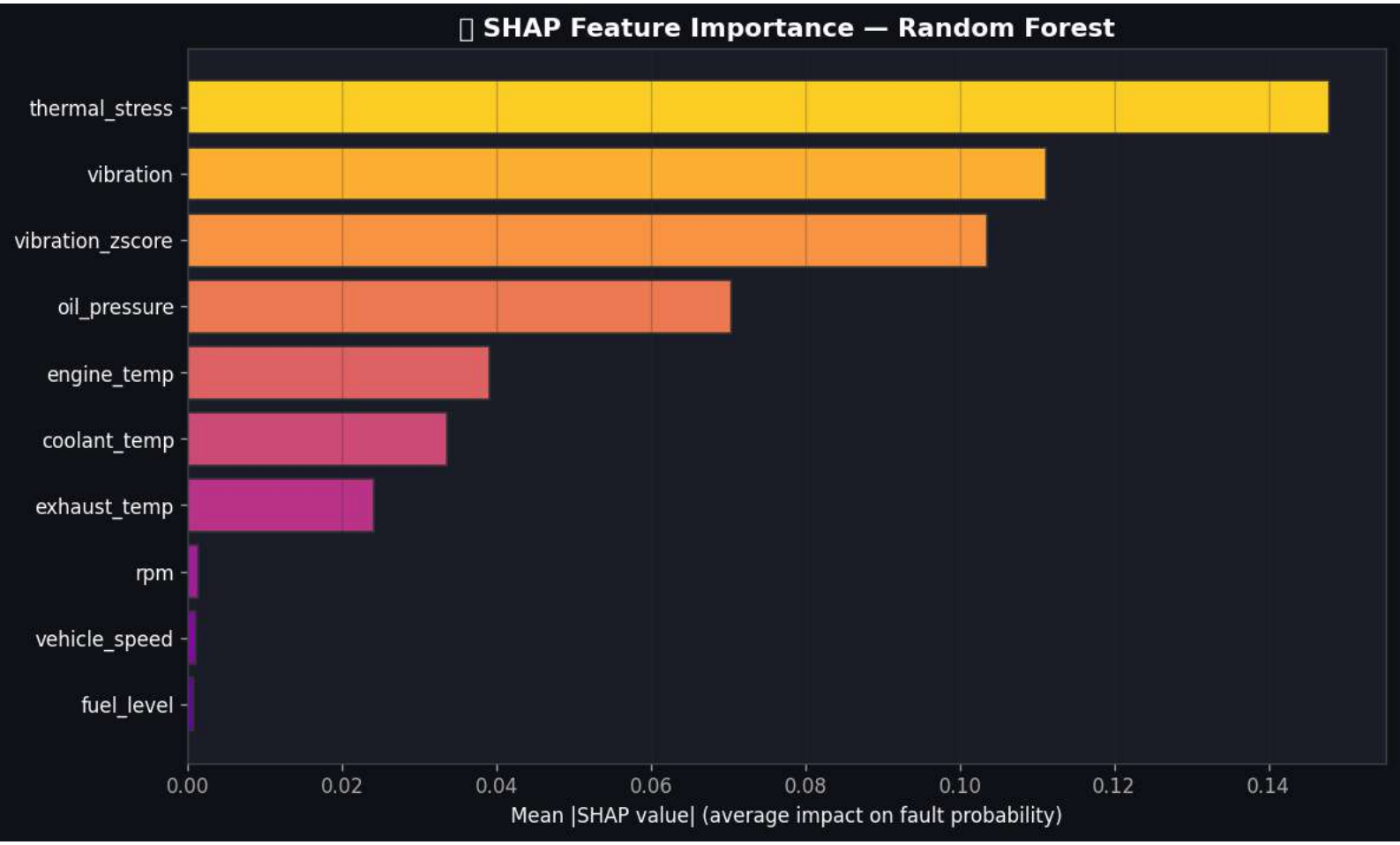
Battery Health — Feature Distributions
voltage
current
temperature
state_of_charge
internal_resist
cycle_count
cell_imbalance
power_output
charge_rate
age_months
Normal
Fault

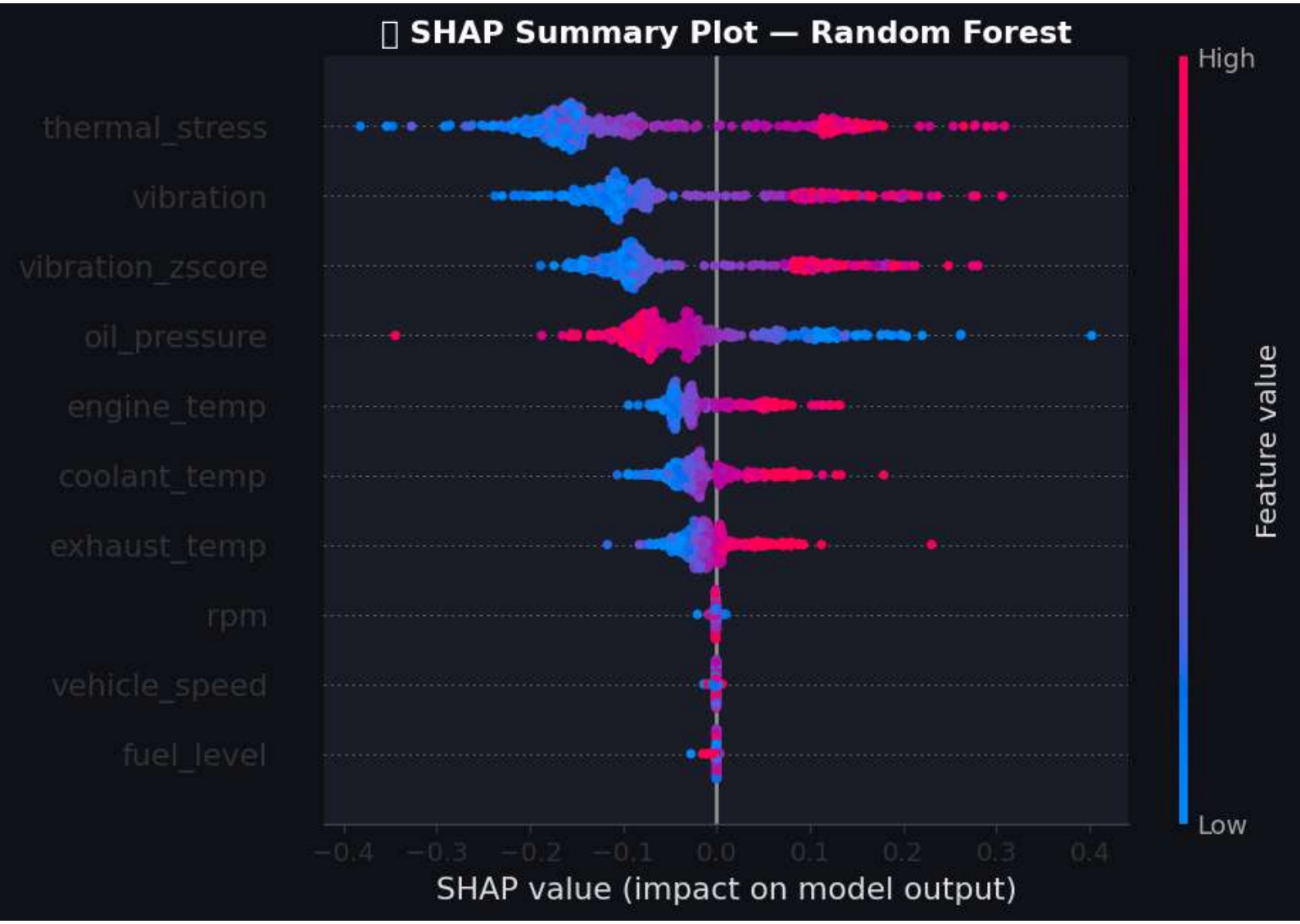

[Vehicle Telemetry] Feature Importance — SelectKBest F-Scores
vibration_zscore
vibration
thermal_stress
engine_temp
oil_pressure
coolant_temp
exhaust_temp
vehicle_speed
rpm
fuel_level
rpm_speed_ratio
brake_pressure
throttle_position
Selection cutoff
0
2000
4000
6000
8000
10000
F-Score (higher = more useful)

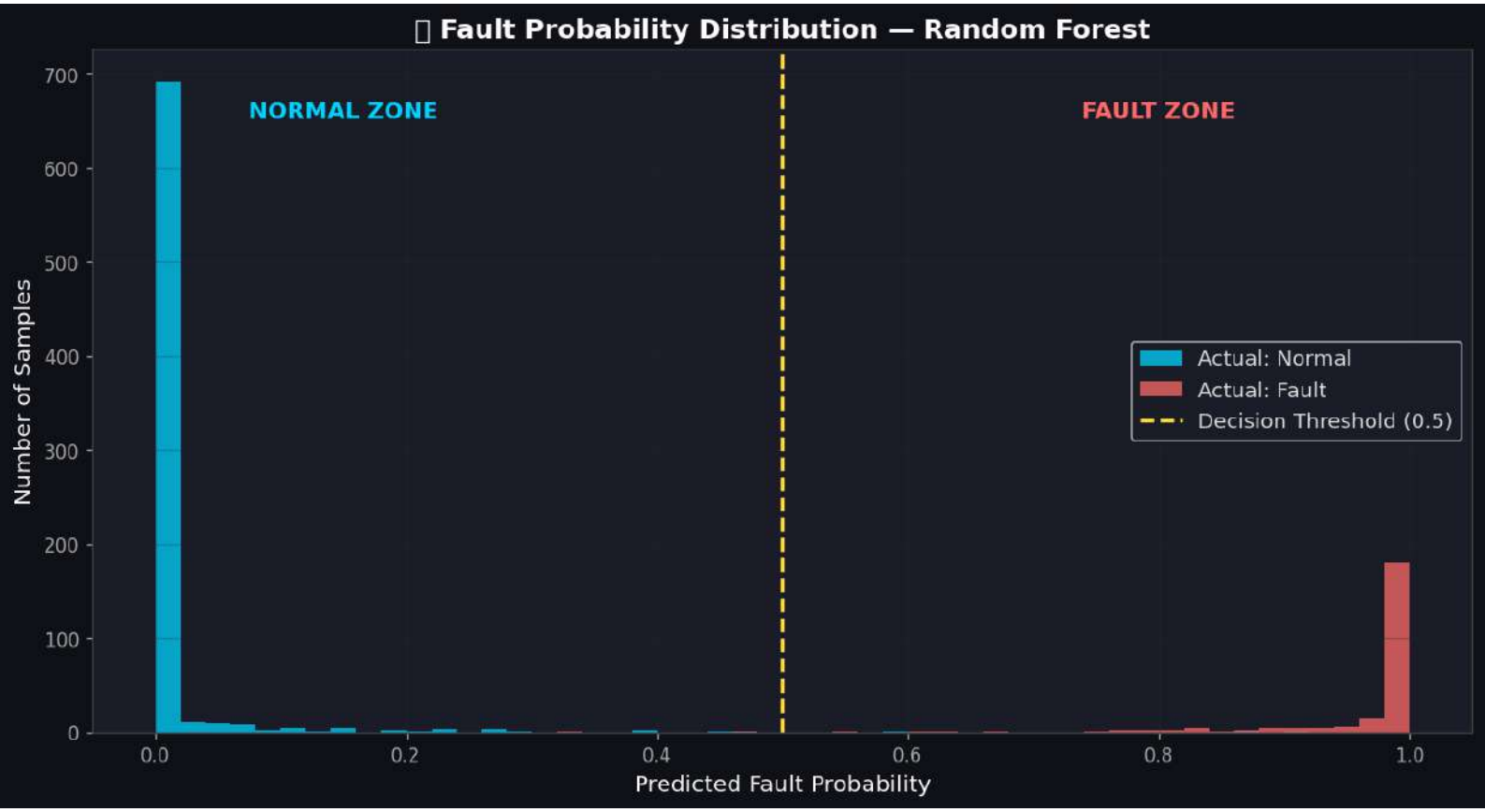

Vehicle Telemetry — Correlation Heatmap
engine_temp
rpm
vehicle_speed
oil_pressure
fuel_level
coolant_temp
throttle_position
brake_pressure
vibration
exhaust_temp
fault
-0.01
-0.02 -0.00
-0.56 0.02 0.02
0.00 0.00 0.00 -0.00
0.55 -0.01 -0.00 -0.53 0.01
-0.01 0.00 0.02 -0.02 -0.00 -0.01
0.00 0.00 -0.01 -0.01 0.00 0.01 0.01
0.63 -0.02 -0.00 -0.63 -0.00 0.60 0.01 0.00
0.48 -0.01 -0.02 -0.47 -0.01 0.46 0.00 0.01 0.52
0.76 -0.02 -0.02 -0.76 -0.00 0.71 0.01 0.00 0.84 0.63
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6


Vehicle Telemetry — Feature Distributions
engine_temp
rpm
vehicle_speed
oil_pressure
fuel_level
coolant_temp
throttle_position
brake_pressure
vibration
exhaust_temp
Normal
Fault

Dataset Class Distribution
Telemetry — Class Balance
Sample Count
4500
4000
3500
3000
2500
2000
1500
1000
500
0
3750
(75.0%)
1250
(25.0%)
Normal (0)
Fault (1)
Battery — Class Balance
Sample Count
2500
2000
1500
1000
500
0
2400
(80.0%)
600
(20.0%)
Normal (0)
Fault (1)


SHAP Feature Importance — Random Forest
thermal_stress
vibration
vibration_zscore
oil_pressure
engine_temp
coolant_temp
exhaust_temp
rpm
vehicle_speed
fuel_level
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Mean |SHAP value| (average impact on fault probability)

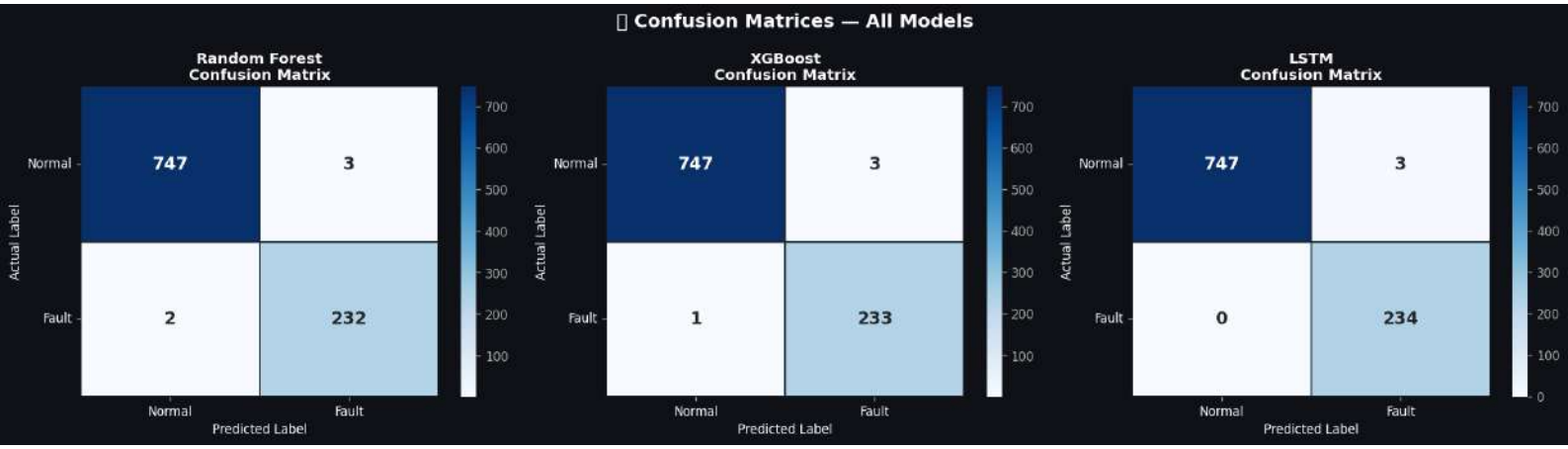

SHAP Summary Plot — Random Forest
thermal_stress
vibration
vibration_zscore
oil_pressure
engine_temp
coolant_temp
exhaust_temp
rpm
vehicle_speed
fuel_level
−0.4
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
SHAP value (impact on model output)
High
Feature value
Low

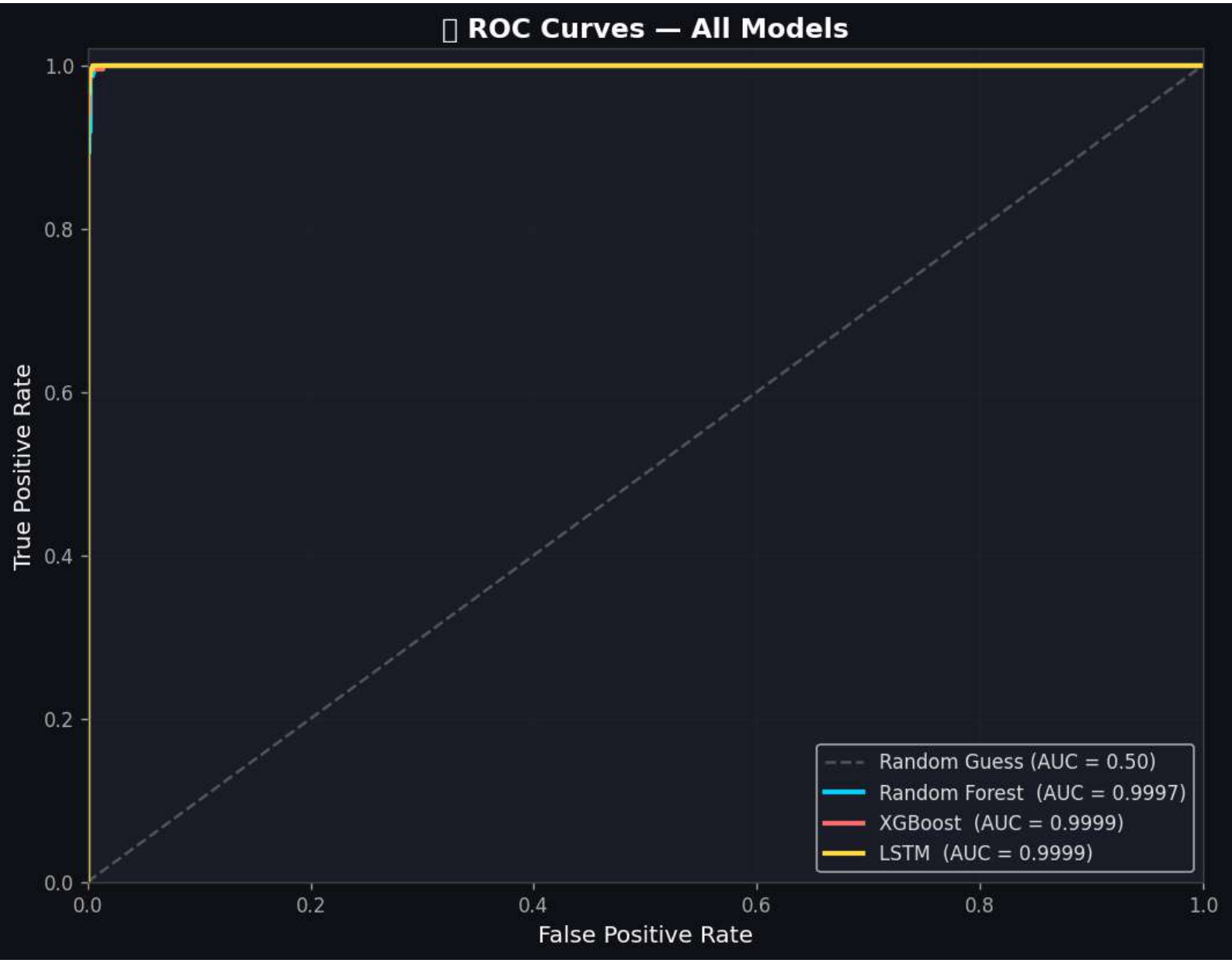

Fault Probability Distribution — Random Forest
NORMAL ZONE
FAULT ZONE
Actual: Normal
Actual: Fault
Decision Threshold (0.5)
Number of Samples
Predicted Fault Probability
0
100
200
300
400
500
600
700
0.0
0.2
0.4
0.6
0.8
1.0

Model Comparison — All Metrics
Score (0–1)
1.0
0.8
0.6
0.4
0.2
0.0
0.995
0.996
0.997
0.987
0.987
0.987
0.992
0.996
1.000
0.989
0.992
0.994
1.000
1.000
1.000
Accuracy
Precision
Recall
F1 Score
AUC-ROC
Random Forest
XGBoost
LSTM

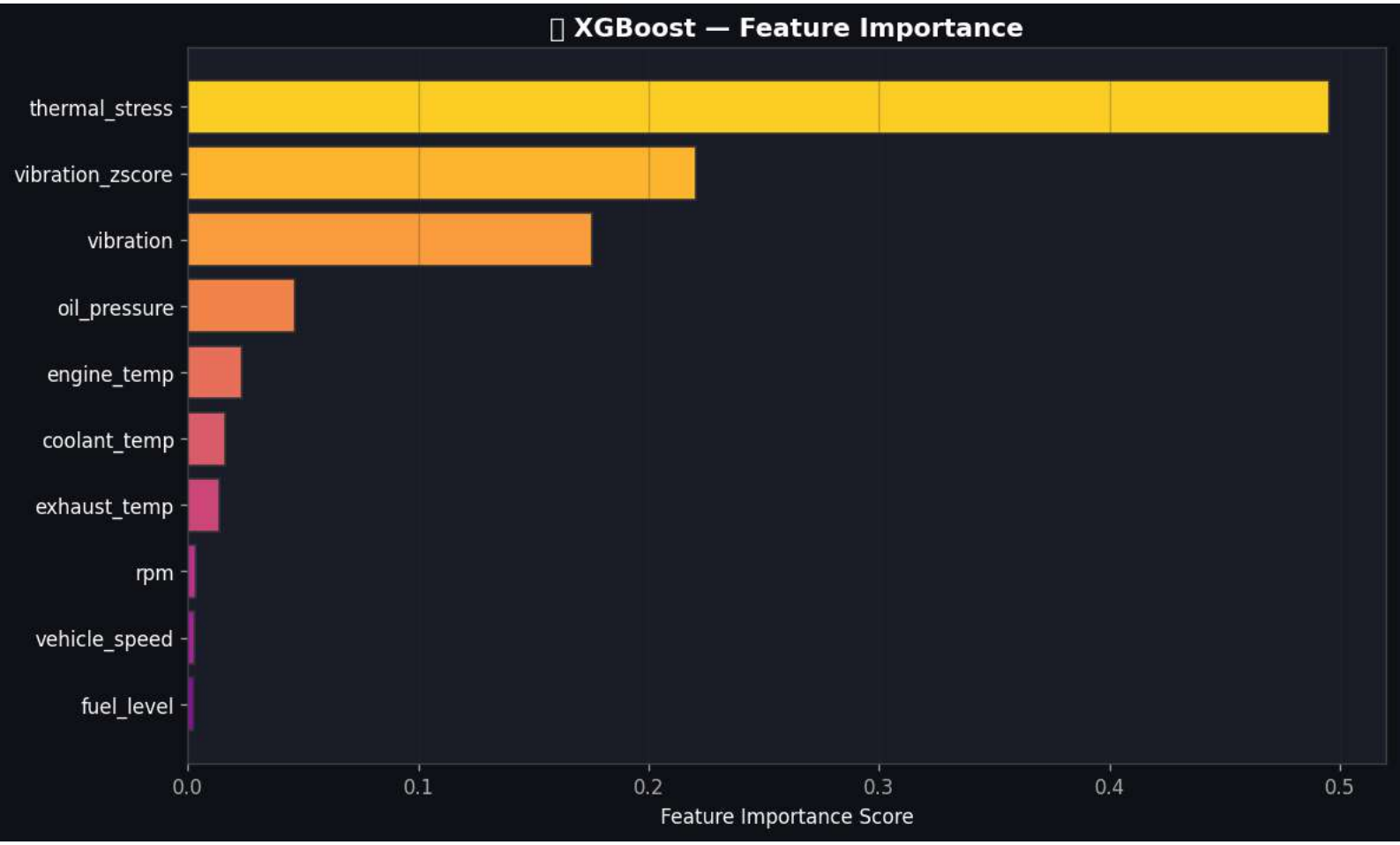
📊 Model Performance Comparison

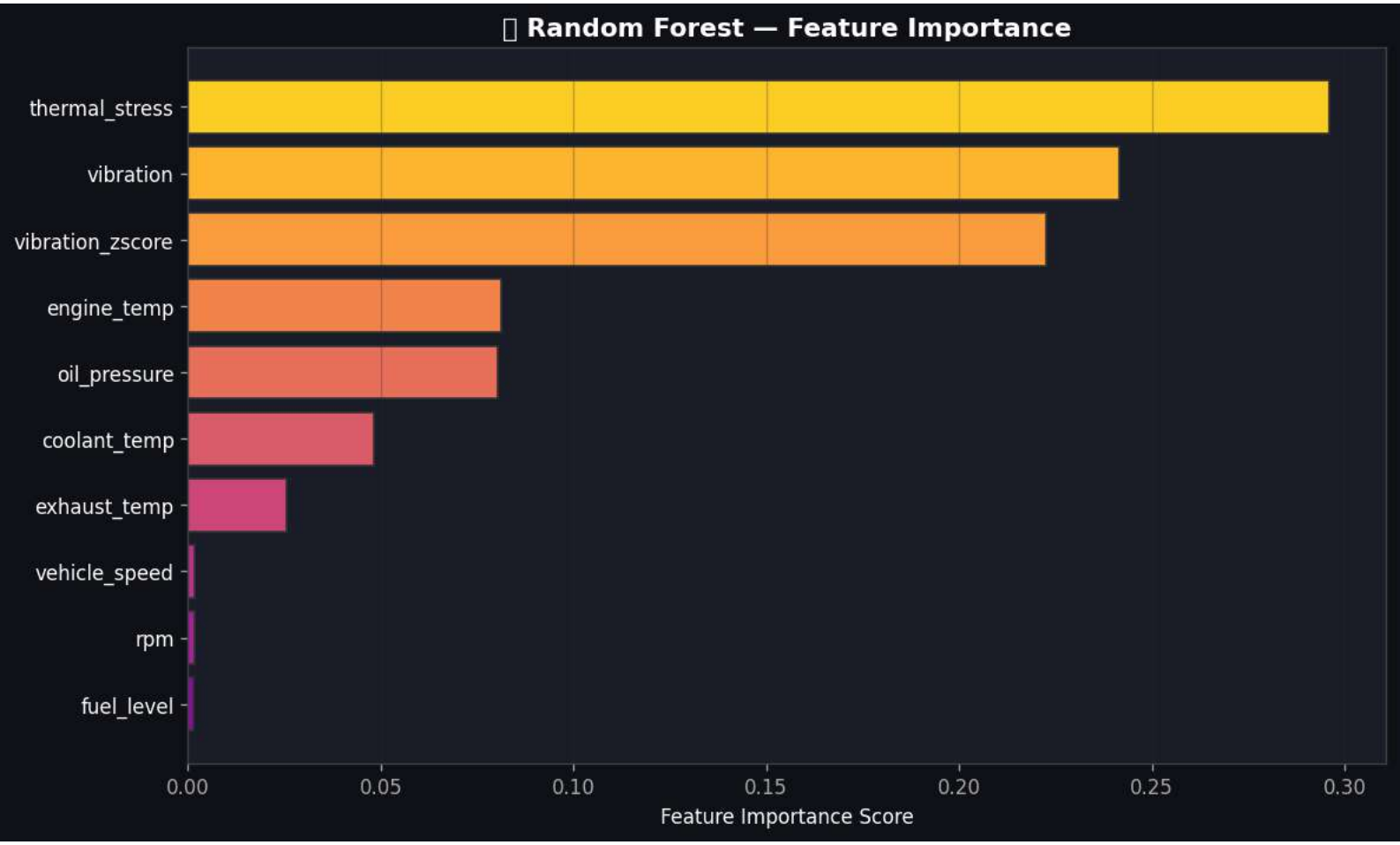
Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC	Train Time (s)
Random Forest	0.9949	0.9872	0.9915	0.9893	0.9997	0.59
XGBoost	0.9959	0.9873	0.9957	0.9915	0.9999	0.57
LSTM	0.9970	0.9873	1.0000	0.9936	0.9999	23.25

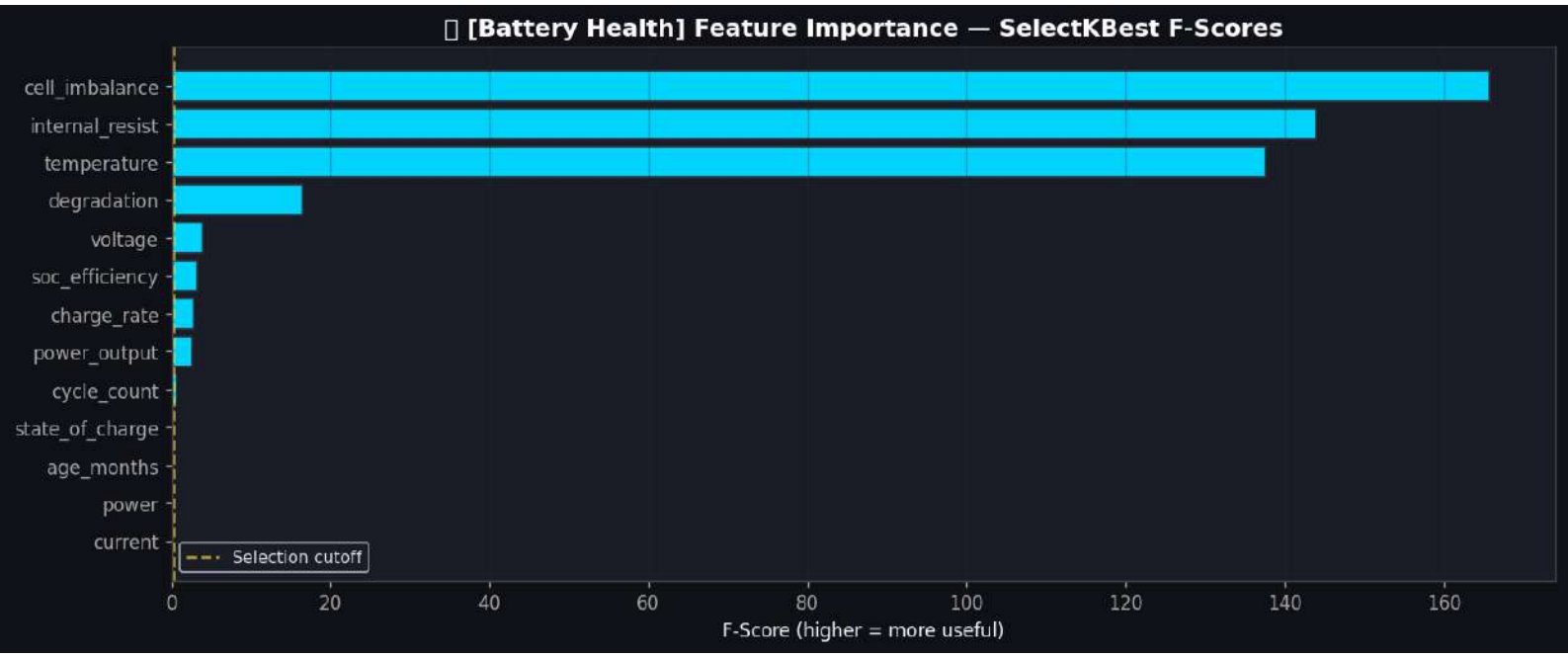

Confusion Matrices — All Models
Random Forest
Confusion Matrix
Normal
Fault
747
3
2
232
Actual Label
Predicted Label
XGBoost
Confusion Matrix
747
3
1
233
LSTM
Confusion Matrix
747
3
0
234
700
600
500
400
300
200
100
0

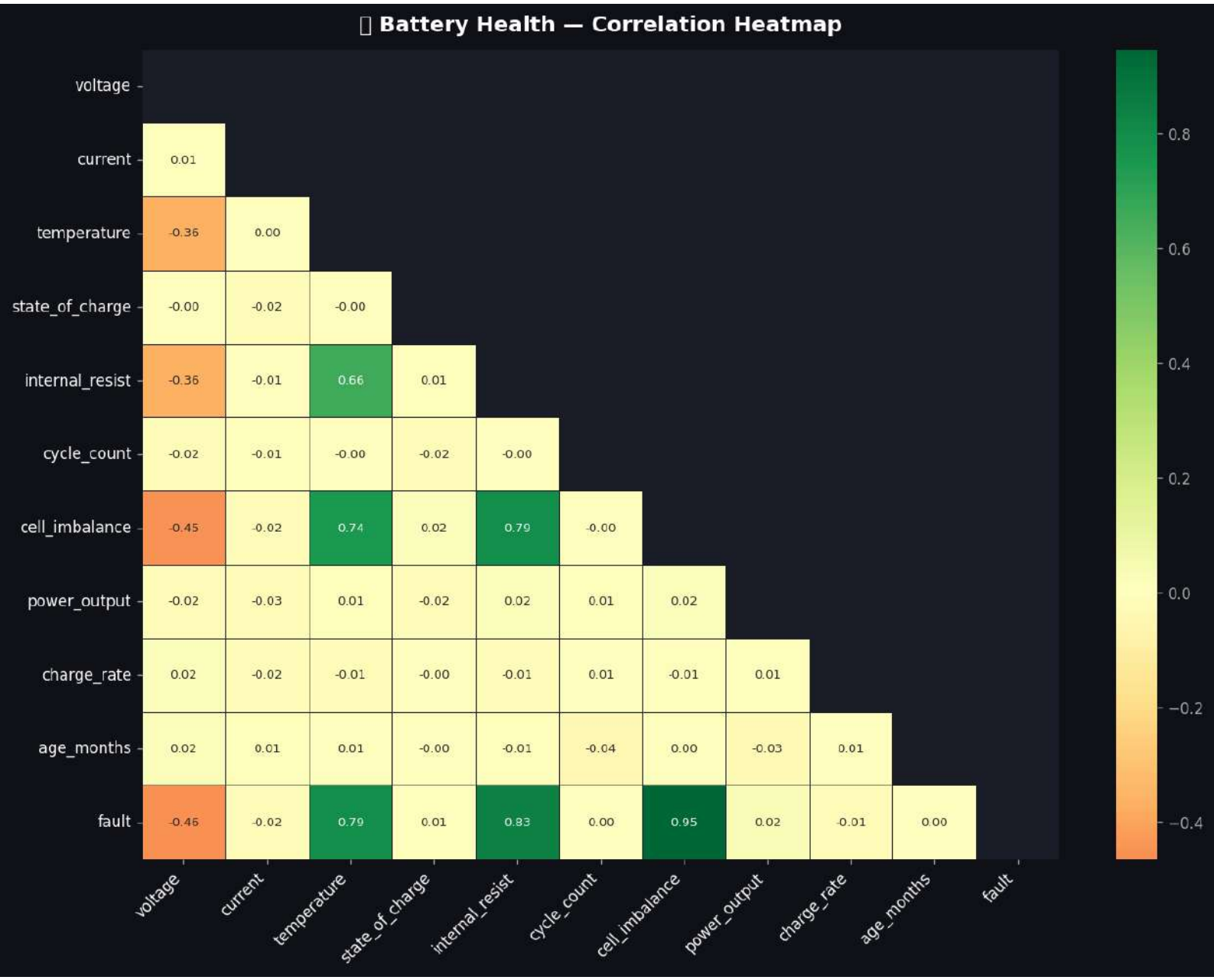

ROC Curves — All Models
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Random Guess (AUC = 0.50)
Random Forest (AUC = 0.9997)
XGBoost (AUC = 0.9999)
LSTM (AUC = 0.9999)

LSTM Training History
Training Loss
Train
Validation
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
5
10
15
20
25
30
Epoch
Training Accuracy
1.000
0.995
0.990
0.985
0.980
0.975
0.970
Train
Validation
0
5
10
15
20
25
30
Epoch


XGBoost — Feature Importance
thermal_stress
vibration_zscore
vibration
oil_pressure
engine_temp
coolant_temp
exhaust_temp
rpm
vehicle_speed
fuel_level
0.0
0.1
0.2
0.3
0.4
0.5
Feature Importance Score


Random Forest — Feature Importance
thermal_stress
vibration
vibration_zscore
engine_temp
oil_pressure
coolant_temp
exhaust_temp
vehicle_speed
rpm
fuel_level
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Feature Importance Score


[Battery Health] Feature Importance — SelectKBest F-Scores
cell_imbalance
internal_resist
temperature
degradation
voltage
soc_efficiency
charge_rate
power_output
cycle_count
state_of_charge
age_months
power
current
Selection cutoff
0
20
40
60
80
100
120
140
160
F-Score (higher = more useful)


Battery Health — Correlation Heatmap
voltage
current
temperature
state_of_charge
internal_resist
cycle_count
cell_imbalance
power_output
charge_rate
age_months
fault
0.01
-0.36 0.00
-0.00 -0.02 -0.00
-0.36 -0.01 0.66 0.01
-0.02 -0.01 -0.00 -0.02 -0.00
-0.45 -0.02 0.74 0.02 0.79 -0.00
-0.02 -0.03 0.01 -0.02 0.02 0.01 0.02
0.02 -0.02 -0.01 -0.00 -0.01 0.01 -0.01 0.01
0.02 0.01 0.01 -0.00 -0.01 -0.04 0.00 -0.03 0.01
-0.46 -0.02 0.79 0.01 0.83 0.00 0.95 0.02 -0.01 0.00
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4